

Design and experiment of an integrated navigation system for a paddy field scouting robot

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ABSTRACT

The navigation system is a crucial part of a paddy field scouting robot, which directly affects the working performance of the robot. The navigation error of the robot was enlarged by the inclination of the robot, which was caused by the uneven surface of the paddy fields. In this study, a map-based navigation system was proposed. The robot navigation map was derived from the planting map of a transplanter. An IMU was integrated with the GNSS to correct the navigation error using the inclination data of the robot through the Kalman filtering technique. A series of simulations and tests were performed to evaluate the performance of the integrated navigation system, including a line following test, a rough road simulation and test, and a row tracking test, as well as a field validation test. The lateral position offset and bearing angle deviation were measured and analyzed to assess the performance. The test results showed that the integrated navigation system significantly reduced the lateral position offset and bearing angle variation by 38.3 % and 26.7 % respectively as compared to the control group without the IMU correction. In the field validation test, the average lateral position offset was 5.93 cm, which was deemed acceptable for paddy field scouting operation in terms of navigation accuracy. In conclusion, the proposed integrated navigation system could be readily applied to the paddy field scouting robot to minimize navigation errors due to uneven ground and random inclination.

1. Introduction

Field scouting is commonly conducted to collect field data for evaluating crop conditions, which is a very important component of precision agriculture (Yamasaki et al., 2022). Frequent and accurate field scouting provides site-specific information on crop health and nutrition, diseases, insects, and weeds in a timely fashion. This information is essential for the determination and implementation of appropriate field management strategies (Fishel et al., 2018). Effective strategies reduce the management area from a whole farm field down to a sub-field level to realize site-specific agriculture. At the same time, most crops are highly sensitive to changes in environmental and physical conditions, which requires that the scouting process has minimal effects on the crops and their surroundings (Eizicovits and Berman, 2014).

Field scouting is achieved through three main methods, i.e., manual,

unmanned aerial vehicles (UAV)-based, and robot-based scouting. Manual scouting is time-consuming and labor-intensive, which is typically limited to being conducted in a small portion of the field. Furthermore, the result of manual scouting is highly affected by the scout rating repeatability and reliability (Kalischuk et al. 2019). The UAVs can collect data for the whole field in a very short time, which significantly reduced working hours and labor demands (Xie and Yang, 2020). However, the aerial images contain overlapping leaves and shadows, which hindered the accuracy of UAV scouting (Zhang et al., 2022). Furthermore, UAV-based scouting is unable to detect weeds and diseases under the crop canopy (Olson and Anderson, 2021). Robotic field scouting is a promising method of effective detection of crop stresses under the canopy. The robots could also provide more accurate crop information thanks to their near-sensing abilities as compared to the UAVs. At the same time, compared with manual scouting, robotic

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applications are more economically feasible in commercial crop productions (Pedersen et al., 2006).

Robots have been successfully used in various agricultural field operations, such as seeding, spraying, weeding, and harvesting (Bechar and Vigenault, 2016). The use of robots for field scouting operations is still a relatively new concept with limited commercially available robots, for example, SentiV of Meropy and Solix of Solinftec were both released in 2023. However, none of the existing products and technologies were developed specifically for paddy field applications. Grimstad et al. (2015) developed a versatile robot for operating in agricultural fields, where all wheels are in contact with the ground at all times to ensure the mobility and steering of the field robot. The authors suggested that agricultural robots should be lightweight to prevent them from getting stuck, especially in wet soil conditions. Meanwhile, the size of the robot could affect the crop and soil structure via compaction in the scouting process (Bengochea-Guevara et al., 2016). Therefore, the physical attributes of robots including their size and weight are usually determined by the working space in relation to plant characteristics and field conditions. The navigation system is a crucial part of the agricultural robot as well. There are two different types of navigation systems, namely single and integrated navigation systems. Single navigation systems utilize one of the technologies, such as GNSS, vision, LiDAR, ultrasonic, and inertial navigation systems (INS), in guiding the robots. Single navigation systems are typically readily available and low in cost but might experience difficulties in providing a robust and accurate positioning solution. Integrated navigation systems typically combine GNSS with other sensors, such as machine vision cameras and inertial measurement units (IMU), to offer a more accurate and precise position by using the complementary characteristics of different sensors (Li et al., 2021). Different navigation systems have been applied in various robots working in a variety of working environments. For example, Underwood et al. (2015) integrated GPS with LiDAR systems for an orchard robot where the strength of satellite signals is often affected by lush branches.

Field scouting robots require advanced technologies to deal with complex and highly variable environments and objects in agricultural fields, as compared to industrial applications (Hiremath et al., 2014). Paddy fields are generally in a muddy state with an irregular soil surface. Rice transplanters and other agricultural machines being used before field scouting operations would evidently leave wheel ruts on the soil surface. The field is kept flooded with stagnant water for rice growth, which exacerbated the effects of field traffic on surface unevenness. Meanwhile, field scouting robots are small in size and light in weight due to narrow row spacings in paddy fields, which is 30 ± 5 cm in most situations (Hu et al., 2020), tending to 'walk' on the surface of the field. The uneven surface would cause frequent random inclination of the robot. On top of that, given the GPS receiver is generally mounted at the highest point of the robot, the inclination would be translated to large offsets between the desired and actual location of the robot, which would lead to relatively high navigation errors.

Different integrated navigation systems have been proposed to improve the navigation accuracy and precision of agricultural robots. The GPS and the machine vision were coupled through a fuzzy logical model in Rovira-mas et al. (2005), where the GPS tracking errors were corrected by a machine vision-based local positioning technique. Bietresato et al. (2016) developed a LiDAR-based 3D-stereoscopic vision system to monitor the canopy volume in a controlled environment, where two LiDAR sensors were aligned vertically to achieve a lateral-linear-stereoscopic vision. A machine vision-ultrasonic sensor fusion system was proposed by Chen et al. (2018), where the camera was used to collect tree trunk images and the ultrasonic sensor was used to reduce the error of mobile robot localization. The integrated navigation systems mentioned above are sensitive to environmental factors such as lighting, weeds, and other background noises, resulting in many practical difficulties in field implementation; whereas the IMU is relatively independent of these external effects (Meng et al., 2017). Previous studies have attempted to integrate the GPS and IMU utilizing and mitigating the strengths and

limitations, where different information of the IMU was used to satisfy the demand of different systems. A multisensory system coupled the data from the GPS and the IMU using the Kalman filter, where the position information was provided by the GPS, and the acceleration was monitored by the IMU (Caron et al., 2006). The same integration technology was adapted for a boat navigation system (Cahyadi et al., 2023) with the difference that the orientation data of IMU was used to calculate the motion velocity of the boat. To the best of the authors' knowledge, there is no study integrating the GNSS and the inclination data obtained through the IMU for the application of paddy field scouting.

The overarching goal of this study was to develop a GNSS-IMU integrated system through the Kalman filtering technique to improve the navigation accuracy and precision of a paddy field scouting robot. The specific objectives were to (1) propose a map-based navigation method where the robot navigation map was derived from the planting map of a transplanter; (2) correct the navigation errors of a field scouting robot due to uneven ground in paddy fields; and (3) evaluate the navigation system performance through a series of simulations and tests.

2. Materials and methods

2.1. Paddy field scouting robot

The robot consists of a body, drive wheels, driven wheels, a lifting device, a battery, motors, transmission systems, and a control system (Fig. 1). Two drive wheels were placed on the sides of the robot body to provide the driving force. Each drive wheel was powered by a brushless DC electric motor (M3508, DJI, China) through a timing belt-based transmission system. The turning action was realized through the differential motion of the drive wheels controlled by two motors separately. Two driven wheels were arranged on the back of the body and connected through a lifting device to assist in turning and posture adjustments. The robot had an overall length of 560 mm, a width of 415 mm, and a height of 490 mm with a ground clearance of 300 mm. The robot weighed approximately 13 kg.

The control system of the robot consists of a microcontroller (STM32F4, STMicroelectronics, Geneva, Switzerland), wireless remote

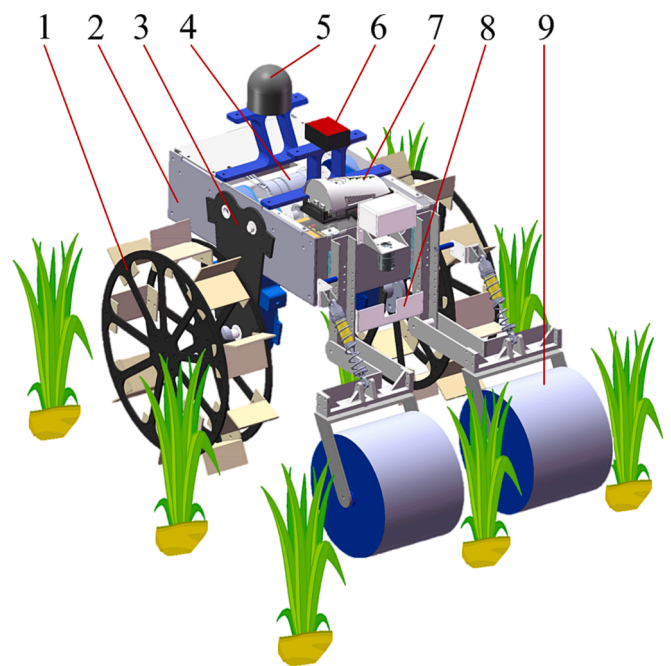


Fig. 1. Schematic diagram of the paddy field scouting robot with principal components (1-drive wheel; 2-body; 3-transmission system; 4-motor; 5-GNSS mobile receiver; 6-IMU; 7-battery; 8-lifting device; 9-driven wheel).

controller, motor controller, inertia measurement unit (IMU), and a navigation system (Fig. 2). The system was supplied with a 24 VDC battery (24v s1, DJI, China). The remote control system includes a wireless remote controller and a remote controller receiver, which is connected to the microcontroller of the robot, which enables manual control and emergency braking. The microcontroller is responsible for collecting and processing data and information from various sensors and controllers. A GNSS receiver was mounted above the body at the leading end of the robot to receive satellite signals. An IMU sensor was placed at the center of the body on the upper surface to monitor the orientation and angular rate of the robot. The pitch angle of the IMU dictated the lifting device control response to minimize the risk of rollover. Meanwhile, the linear accelerations of the IMU were used to derive the actual translational velocities, which were compared with the target velocities to eliminate the differences by adjusting the output power of the motors. A CAN-compatible field-oriented controller (C610, DJI, China) was selected for the DC motors to enable a closed-loop control system for the stability of the robot. The closed-loop control system of the motors also ensures the adaptability of the robot in complex paddy field conditions. The control system was developed based on the CAN BUS communication protocol.

2.2. Integrated navigation system

The proposed navigation system incorporated the GNSS navigation with the IMU-based navigation mechanism. The system consists of a navigation microcontroller (STM32F1, STMicroelectronics, Geneva, Switzerland), a GNSS mobile receiver (Top502, TopGNSS, China), a GNSS base station (Top100, TopGNSS, China), and an IMU (HWT905-TTL, Wit-motion, China) (Fig. 3). The transplanter planting map was being processed via a variety of projection and route planning techniques to generate the target navigation map for the robot, which will be described in detail in Section 2.2.1. The GNSS mobile receiver compared the signals from satellites and the base station to realize RTK, providing up to centimeter-level accuracy in the real-time position information of the robot. The positioning information and the orientation and angular rate information obtained through the IMU were used to develop a kinematics model of the robot, which will be explained in Section 2.2.2. The kinematics model of the robot was then solved using the Kalman filtering method to generate actual position information, which will be reported in Section 2.2.3. The actual position information was compared with the target navigation map at a predetermined timestep using the microcontroller of the control system to actuate the drive wheel motors in Fig. 2.

2.2.1. Map-based navigation

A map-based navigation method was proposed to derive the navigation map for the robot from the planting map of the transplanter (Fig. 4a). A GNSS receiver was installed at the center of the transplanter (PZ-60, Iseki, Japan), which had a wheelbase of 1000 mm and a track

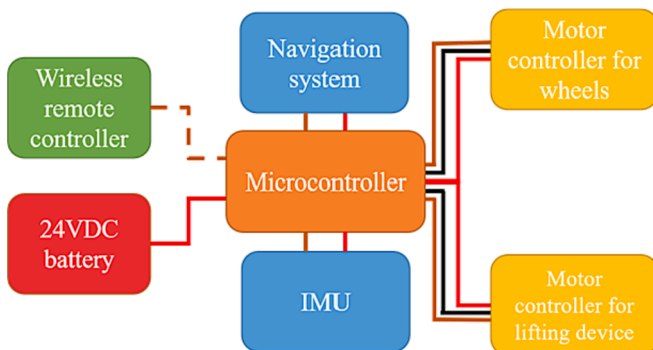


Fig. 2. The control system of the robot.

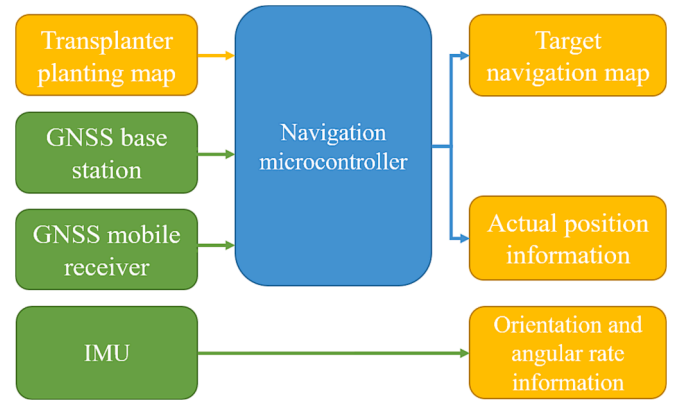


Fig. 3. The integrated navigation system of the robot.

width of 1200 mm with a plant row spacing of 300 mm. An as-planted map of the transplanter was generated as a series of densely distributed points showing real-time locations of the transplanter, which is shown as the red dotted line in Fig. 4a. The position of rice seedlings is presented by the green lines in the Fig. 4a. The planting map needs to be properly discretized before it can be used to produce the navigation map for the robot (Fig. 4b). On the one hand, a bulky map would significantly increase the computational power demand of the robot control system. On the other hand, an overly scattered map would compromise the accuracy of the map by losing important path features. Therefore, an appropriate distance between adjacent points of the map was calculated based on the following equation.

$$D = 2 * \sqrt{\left(R + \frac{L}{2}\right)^2 - R^2} \quad (1)$$

Where D is the distance between adjacent points of the discretized map; R is the distance between the center of the minimum turning circle and the outermost row of the seedlings; L is the row spacing.

The spherical law of cosines equations was used to determine the distance between any two points of the planting map. The discretized planting map was generated by only including the points spaced at the calculated distance with a known starting point.

To improve the estimation accuracy of the navigation map, a Gauss-Kruger projection procedure was applied to convert latitude and longitude information to planar coordinates of the points on the planting map (Fig. 4c). The coordinates of the navigation map points were derived using the following equations.

$$x_i = x_0 + d_i * \cos(\theta) \quad (2)$$

$$y_i = y_0 + d_i * \sin(\theta) \quad (3)$$

Where x_i and y_i are the coordinates of the navigation map points; x_0 and y_0 are the coordinates of the planting map points; d_i is the corresponding distance between the points based on the row spacing distance of the transplanter; θ is the bearing angle of the points; x is in the North direction.

Finally, the coordinates of the navigation map points were used to generate the latitude and longitude based map based on a reversed Gauss-Kruger projection procedure. Fig. 4d shows the navigation map for the robot by connecting adjacent points with straight lines.

2.2.2. Robot kinematic model development

To develop a kinematic model of the robot, a kinematic analysis was conducted to quantitatively describe the motion dynamics and state of the robot (Fig. 5). In the analysis, the following simplifications and assumptions were made to the model: only drive wheels were considered by ignoring the supporting driven wheels; no deformation occurred between the drive wheels and paddy soil; slip was small enough to be

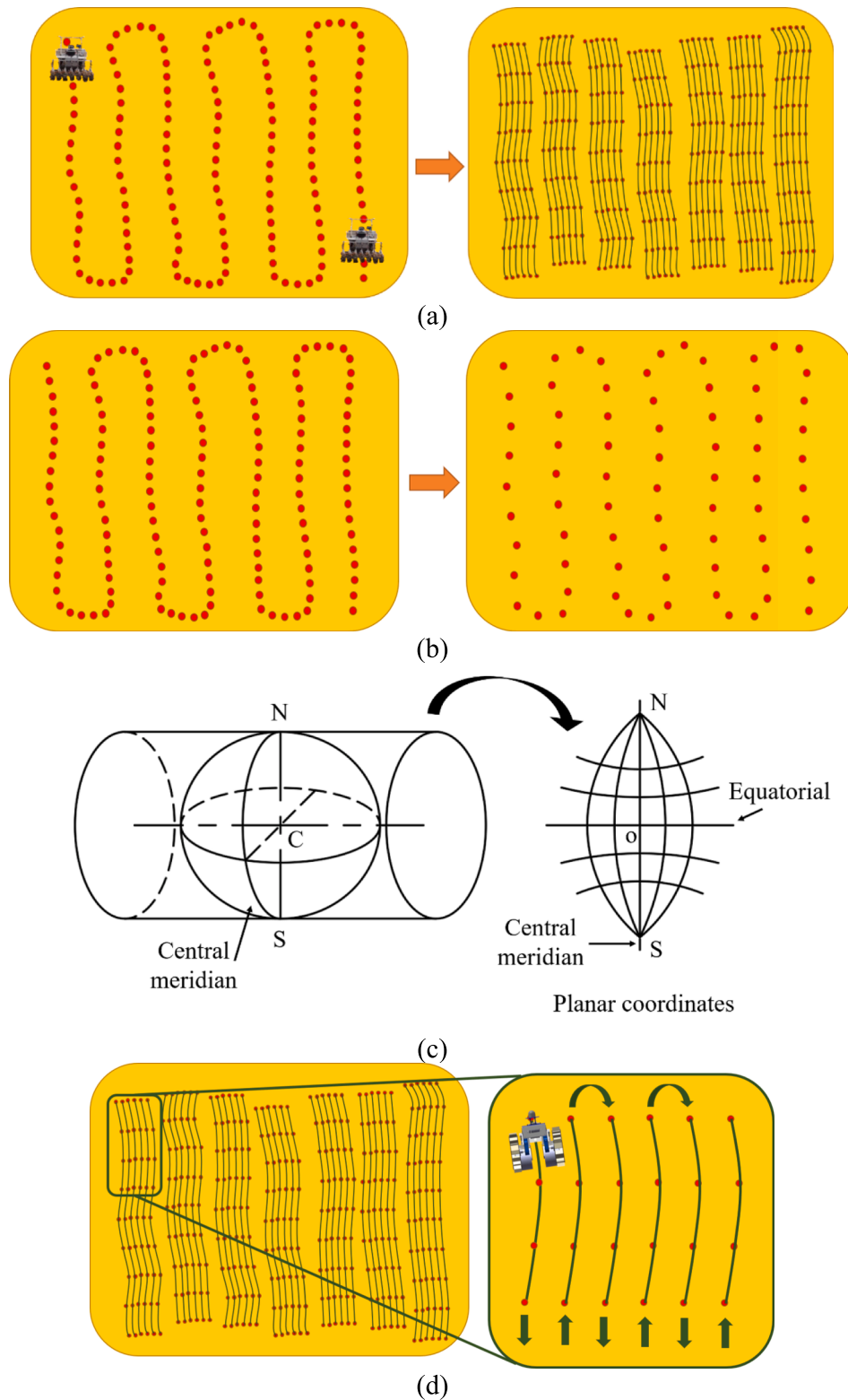


Fig. 4. The map-based navigation system of the robot: (a) from transplanter planting map to robot navigation map; (b) discretization of the planting map; (c) Gauss-Kruger projection; and (d) enlarged view of the robot navigation map.

neglected. Considering the rough terrain of paddy fields, the robot of typical inclination status is shown in Fig. 5a. Projecting the robot to the horizontal plane, one could take the center point of the drive wheels as the reference point (P) of the robot (Fig. 5b). The location and orientation could be described by x coordinate (P_x), y coordinate (P_y), and bearing angle (θ). The linear velocity (v_c), which is the average of the

linear velocities of the drive wheels, v_l and v_r , dictated the planar motion in conjunction with the angular velocity (ω_1). The inclination resulted in the shift of the reference point on the vertical plane, which could be compensated by including the inclination angle (α) and inclination angular velocity (ω_2) in the kinematic model (Fig. 5c). Therefore, the state vector (X) could be written as the following equation to depict the

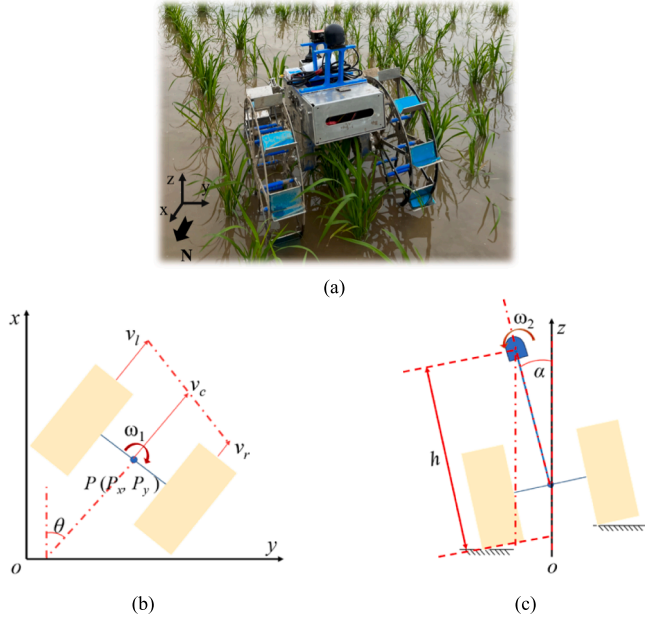


Fig. 5. The kinematic analysis of the robot: (a) a typical motion status of the robot with inclination; (b) the top view of the motion dynamics model; and (c) the front view of the motion dynamics model.

real-time motion dynamics of the robot.

$$X = [P_y \ P_x \ v_c \ \theta \ \omega_1 \ \alpha \ \omega_2]^T \quad (4)$$

2.2.3. Kalman filtering

To obtain more accurate motion dynamics and the state of the robot, Kalman filtering was used on the state vector X . The filtered state vector provided the true values of the real-time position information of the robot. The Kalman filtering consists of three stages, i.e., prediction, measurement, and update, for each time step. The prediction stage is responsible for predicting the state vector (X'_k) and state covariance matrix (P'_k) of the current moment based on the information of the previous time step using equations (5)–(7). The measurement stage compares the measured data and the predicted values to yield innovation residual (Y), innovation covariance matrix (S), and Kalman gain (K) using equations (8)–(11). The update stage focuses on obtaining the final state vector and state covariance matrix based on the calculated Kalman gain using equations (12)–(13). The final information was used as the reference values of the current state and was also used in the prediction stage of the next time step.

At the prediction stage, the state vector of the current time step can be predicted using the following equation.

$$X'_k = F * X_{k-1} + u \quad (5)$$

Where X is the state vector and subscripts indicate the time step; F is a state transition matrix; u is a control vector.

Therefore, equation (5) can be expressed as equation (6) using the installation height of the GNSS mobile receiver (h in Fig. 5c).

$$\begin{bmatrix} P_y \\ P_x \\ v_c \\ \theta \\ \omega_1 \\ \alpha \\ \omega_2 \end{bmatrix}'_k = \begin{bmatrix} 1 & 0 & \sin\theta*\Delta t & 0 & 0 & 0 & 0 \\ 0 & 1 & \cos\theta*\Delta t & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \Delta t & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} P_y \\ P_x \\ v_c \\ \theta \\ \omega_1 \\ \alpha \\ \omega_2 \end{bmatrix}_{k-1} + \begin{bmatrix} h*\sin\alpha*\cos\theta \\ -h*\sin\alpha*\sin\theta \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (6)$$

The state covariance matrix of the current time step can be predicted using the equation below.

$$P'_k = F * P_{k-1} * F^T + Q \quad (7)$$

Where P is the state covariance matrix and subscripts indicate the time step; Q is a process covariance matrix determined through preliminary simulations.

At the measurement stage, the innovation residual was calculated using the following equation to represent the difference between the observed and predicted values.

$$Y = Z - H * X'_k \quad (8)$$

Where Y is the innovation residual; Z is the observation data; H is the measurement matrix.

Therefore, equation (8) can be rewritten as equation (9).

$$\begin{bmatrix} \Delta P_y \\ \Delta P_x \\ \Delta \theta \\ \Delta \alpha \end{bmatrix} = \begin{bmatrix} (P_y)_z \\ (P_x)_z \\ (\theta)_z \\ (\alpha)_z \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} * \begin{bmatrix} P_y \\ P_x \\ v_c \\ \theta \\ \omega_1 \\ \alpha \\ \omega_2 \end{bmatrix}_k \quad (9)$$

The innovation covariance matrix can be calculated using the equation below.

$$S = H * P'_k * H^T + R \quad (10)$$

Where S is the innovation covariance matrix; R is a measurement covariance matrix determined based on sensor specifications.

The Kalman gain can be calculated using the following equation.

$$K = P'_k * H^T * S^{-1} \quad (11)$$

At the update stage, the updated state vector and state covariance matrix can be calculated using the following equations.

$$X_k = X'_k + K * Y \quad (12)$$

$$P_k = (I - K * H) * P'_k \quad (13)$$

Where X_k is the updated state vector; P_k is the updated state covariance matrix; I is an identity matrix of size 7.

2.3. Simulations and experiments

A series of simulations and tests were performed to evaluate the performance of the integrated navigation system of the robot in terms of its accuracy and precision. Firstly, the robot was tested on leveled concrete for its ability to follow a straight line, which is the simplest navigation map. The system's ability to handle frequent ups and downs and minimize the effects of inclination on navigation accuracy and precision through the Kalman filtering technique was simulated and further tested with a rough road test. After that, a row tracking test was conducted in a typical paddy field to examine the navigation system performance in real-world conditions. Finally, a field validation test was carried out using the planting map of the transplanter as the input for the integrated navigation system to comprehensively evaluate the navigation system performance in fields. The lateral position offset and bearing angle deviation were measured and analyzed to evaluate the performance. The robot was operated at a constant velocity of 0.5 m/s in the experiments.

2.3.1. Line following test

The line following test was conducted in an open area with strong satellite signal coverage on concrete. The longitude and latitude information of two random points (A and B) at a distance of 10 m were recorded with the GNSS mobile receiver on the robot (Fig. 6). The Gauss-Kruger projection was used to convert the latitude and longitude data to planar coordinates of the points, i.e., (x_A, y_A) and (x_B, y_B). A straight line

connecting the two points with a format of $ax+by+c=0$ was then derived using the coordinate information. The generated line equation was then sent to the robot to function as the target navigation map. The robot was placed at the starting point A and set to move toward point B by following the target line autonomously. The line following test was repeated 4 times.

2.3.2. Rough road simulation and test

The effectiveness of Kalman filtering in improving navigation performance was first simulated in Matlab 2020a (MathWorks Inc., Natick, MA, USA) and then verified with a rough road test. In the simulation, a series of random numbers following normal distribution $X \sim N(0, 2^2)$ and $X \sim N(0, 3^2)$ were generated as angular velocities and inclination angular velocities respectively to resemble the field working condition on a random uneven ground. The robot had an initial bearing angle of 25° , an initial inclination angle of 0° and an initial linear velocity of 0.5 m/s. The simulations with Kalman filtering were considered as the treatment group and the simulations without Kalman filtering were included as the control group. Each simulation lasted for 500 s and each group had 500 simulation runs. In the rough road test, several rubber speed bumps with heights ranging from 3 to 9 cm were randomly placed at the sides of a straight line with a length of 20 m to construct a rough road course (Fig. 7). The speed bumps were rearranged for each test run. The robot was guided along the rough road course using the navigation system with and without Kalman filtering treatment. The rough road test was repeated 6 times.

2.3.3. Row tracking test

The robot with the verified integrated navigation system was further tested in a typical paddy field (Fig. 8). The field was located in Shapu Farms in Zhaoqin City, Guangdong Province, China (23.14 N and 112.66 E). The field was planted using a modified transplanter (PZ-60, Iseki, Japan) with an automatic guidance system at a row spacing of 0.3 m and a plant spacing of 0.15 m. On the 13th day after the transplanting, seedlings had an average height of 0.2 m. The surface water layer and plow layer had a thickness of approximately 0.1 m each at the testing area. Three random rows of rice seedlings with a high level of straightness were selected with a length of 10 m each. The robot was set to track the straight rows in a manner similar to the line following test discussed above.

2.3.4. Field validation test

The field validation test was carried out in the same field as the row track test. Three random segments of the transplanter planting map, each with a length of 10 m, were extracted and loaded on the robot. The robot was placed near the starting corner of the navigation map and was completely guided by the integrated navigation system to finish the scouting of the six rows within each planting map segment (Fig. 9).

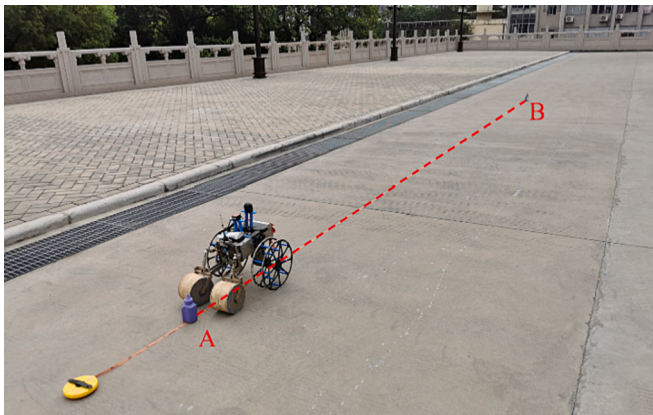


Fig. 6. The line following test.

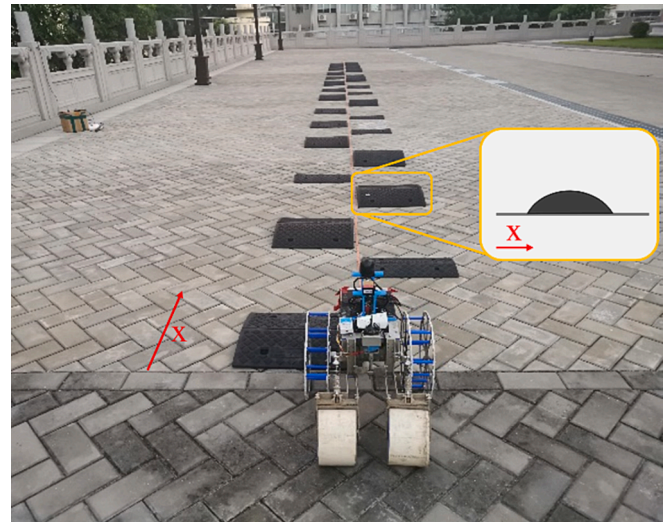


Fig. 7. The rough road test with an enlarged view of the cross-section of a speed bump.



Fig. 8. The row tracking test in a paddy field.

2.3.5. Measurements and data analysis

The performance of the navigation system was evaluated by the lateral position offset and bearing angle deviation. The state vector was sampled at a frequency of 10 Hz to obtain the real-time position coordinates and current bearing angle information.

The lateral position offset could be calculated by equation (14).

$$D_{ij} = \frac{ax_{ij} + by_{ij} + c}{\sqrt{a^2 + b^2}} \quad (14)$$

Where (x_{ij}, y_{ij}) is the real-time position information of the robot; i is the test run number; j indicates sampling points.

The standard deviation of the lateral position offset was calculated by equation (15).

$$\sigma_i = \sqrt{\frac{1}{N} \sum_{j=1}^N (D_{ij} - \bar{D})^2} \quad (15)$$

The bearing angle deviation could be calculated by equation (16).

$$\theta_i = \arctan\left(\frac{y_B - y_A}{x_B - x_A}\right) \quad (16)$$

Where (x_A, y_A) and (x_B, y_B) are the position information of the target



Fig. 9. The field validation test.

points; i is the test run number.

The standard deviation of the bearing angle deviation was calculated by equation (17).

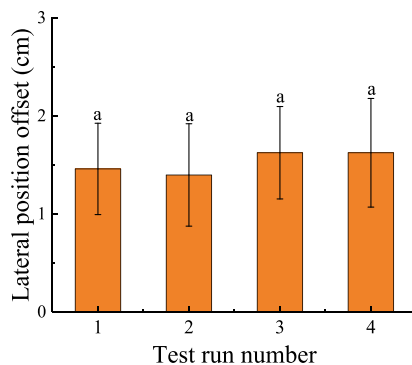
$$\sigma_i = \sqrt{\frac{1}{N} \sum_{j=1}^N (\theta_{ij} - \bar{\theta}_i)^2} \quad (17)$$

ANOVA was performed using statistical software (SPSS v 26.0) to examine the effects of the proposed technique on the accuracy and precision of the robot navigation regarding the position and bearing angle. The means of measured data between test runs and groups were compared with Tukey's multiple range test at the significance level of 0.05.

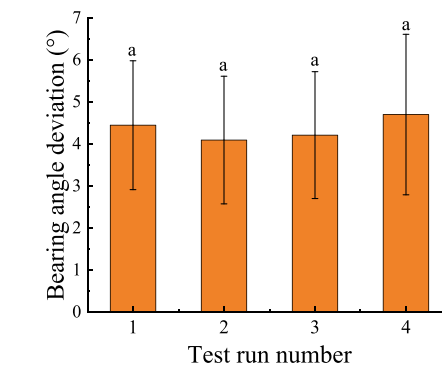
3. Results and discussion

3.1. Line following test results

The robot had an average lateral position offset of 1.53 cm and an average bearing angle deviation of 4.36° in the line following test on a leveled concrete surface (Fig. 10). The small position offset indicated that the robot mechanical system was sound and the map-based navigation system was able to achieve a centimeter-grade positioning accuracy. The lateral position offset was similar to that of Li et al. (2021), whose robot was tested with the same setup. The relatively small bearing angle deviation suggested that the robot was able to self-correct its heading in a comparatively smooth fashion. The lateral position offset varied from 1.40 to 1.62 cm and the bearing angle deviation changed from 4.10° to 4.70° among the 4 test runs. The statistical analysis suggested that there were no significant differences in terms of both the



(a)



(b)

Fig. 10. Lateral position offset (a) and bearing angle deviation (b) of different test runs. Means followed by different letters are significantly different according to Tukey's multiple range test at the significance level of 0.05, and error bars are standard deviations.

lateral position offset and bearing angle deviation. The results indicated that the positioning precision was adequate and the overall operation of the robot was satisfactory on the leveled ground for further prototyping and testing.

3.2. Rough road simulation and test results

Fig. 11 demonstrated that the average lateral position offset of simulations fluctuated around constant values with relatively small variations. The treatment group with Kalman filtering experienced an average position offset of 5.02 cm, which was smaller than that of the control group at 36.2 %. In other words, the Kalman filtering technique improved the navigation accuracy by more than one-third in terms of the lateral position offset. The simulations allowed one to conduct a vast number of test runs with input field condition values following normal distributions without background noises. This offered the benefit of singling out the effect of the Kalman filter on navigation accuracy performance with a sufficient sample size (500 in this case) at a low cost as compared to physical field tests, which helped us to assess the true performance gain due to the added Kalman filtering procedure in the integrated navigation system.

The rough road test results were generally in agreement with the simulation results as shown in Fig. 12. The treatment group with Kalman filtering significantly reduced the lateral position offset from 6.0 to 3.7

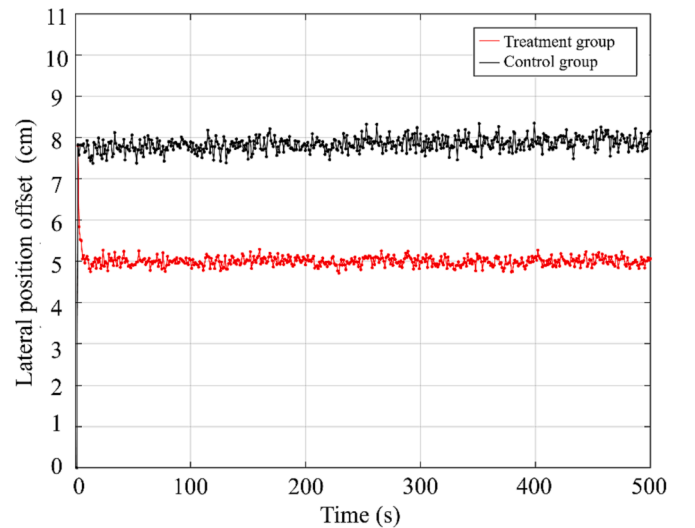


Fig. 11. Simulated lateral position offset of the treatment group with Kalman filtering and of the control group.

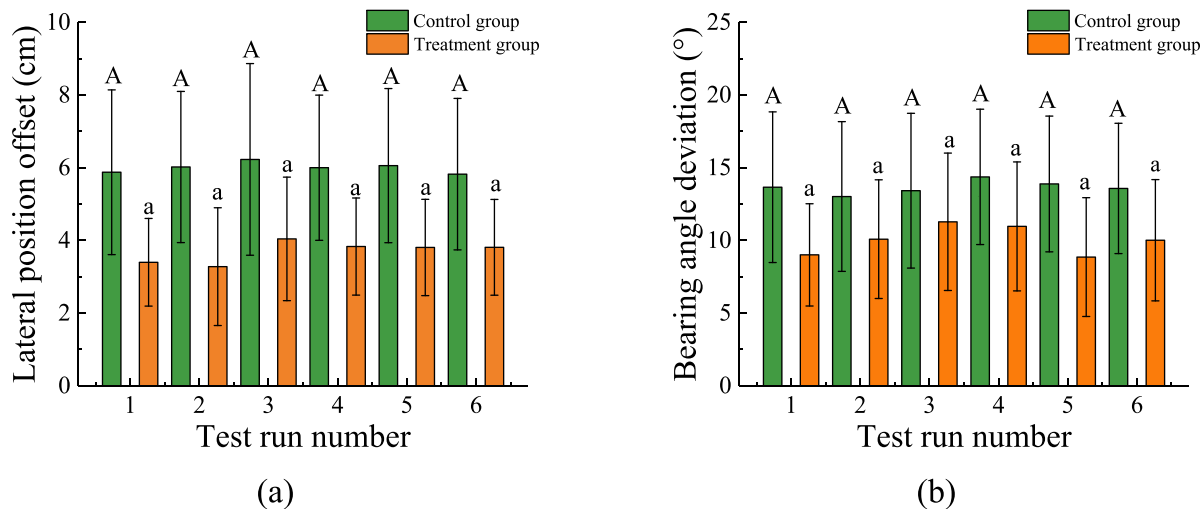


Fig. 12. Lateral position offset (a) and bearing angle deviation (b) of different test runs. Means followed by different letters are significantly different according to Tukey's multiple range test at the significance level of 0.05, and error bars are standard deviations.

cm with a 38.3 % improvement as compared with the control group; similarly, the bearing angle variation was significantly decreased by 26.7 % from 13.7° to 10.0° as compared between the treatment group and the control group. The lateral position offset was lower than that in the experiment carried out by Nagasaka et al. (2004), where a single navigation system was used. The method of using the Kalman filter to incorporate the robot's inclination information into the navigation system effectively reduced the navigation error. The magnitude of navigation accuracy improvement on the lateral position offset was similar in the simulation and test (38.3 % vs 36.2 %), which proved that the Kalman filtering technique was effective in improving the navigation accuracy of the robot. However, the average lateral position offset was smaller in the test as compared to that in the simulation either for the control or treatment group, which was possibly attributed to a more conservative rough road condition used in the test in terms of the average inclination angle of the robot. Nevertheless, the average lateral position offset in both the rough road simulation and test was much larger than that in the line following test, specifically, a 142 % increase in the test and a 228 % increase in the simulation, which illustrated that the navigation accuracy was dominated by the inclination angle variation in this study. Furthermore, the statistical analysis showed that the differences between the test runs within each group were not significant

regardless of the measurements, which suggested that the robot navigation system had a high level of precision under rough road conditions. Li et al. (2021) also tested the effect of Kalman filtering on the navigation accuracy on a smooth surface. The results showed that the average error was reduced by 8.21 % across various experimental conditions, which suggested that the effect of Kalman filtering was more dominant in rough road conditions.

3.3. Row tracking test results

The robot had an average lateral position offset of 2.54 cm and an average bearing angle deviation of 7.78° in the row tracking test in a typical paddy field condition (Fig. 13). Comparing with the rough road test results, the average offset and deviation were reduced in the field condition, which indicated that the experimental paddy field had a modest average roughness. On the other hand, the field showed a significant level of variation in roughness from spots to spots. For example, test run #2 had an average lateral position offset of 3.08 cm, which was significantly higher than that of 2.21 cm measured in test run #3. It was interesting to note that the lateral position offset had an inverse relationship with the bearing angle deviation, i.e., the higher the offset, the lower the deviation, which further manifested the high variation and

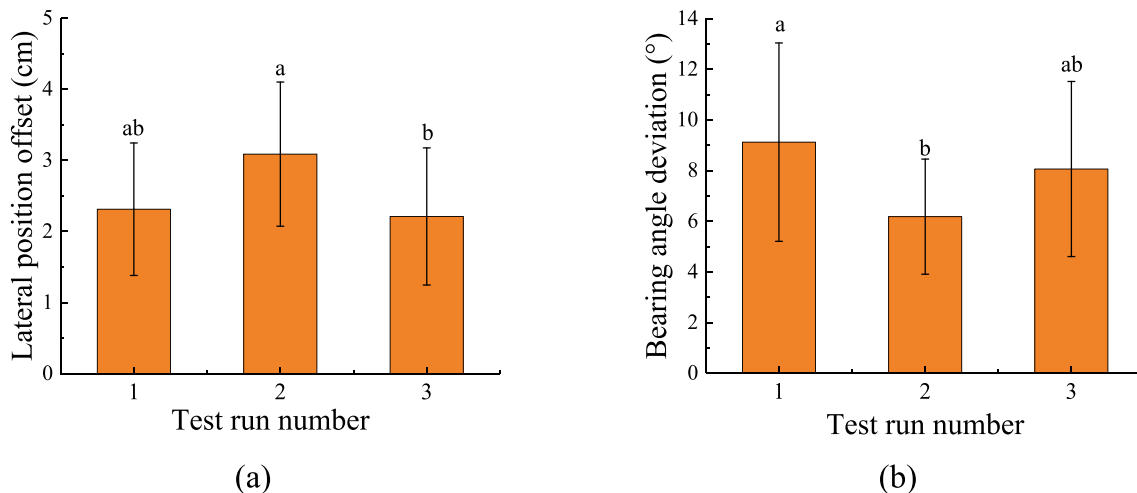


Fig. 13. Lateral position offset (a) and bearing angle deviation (b) of different test runs. Means followed by different letters are significantly different according to Tukey's multiple range test at the significance level of 0.05, and error bars are standard deviations.

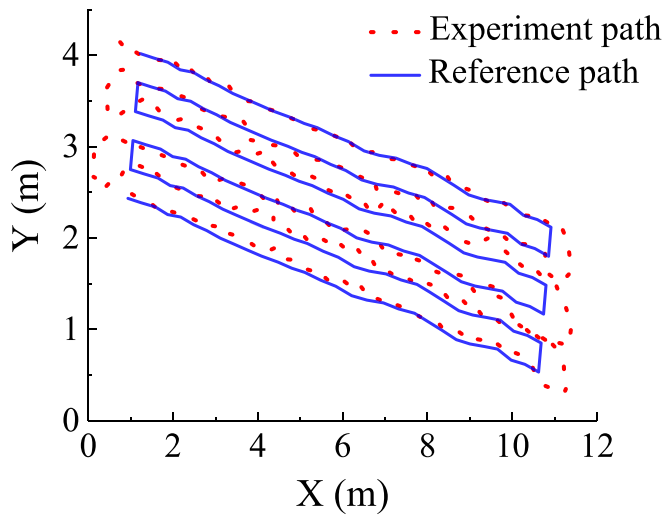


Fig. 14. Comparison between the experiment path and reference path in a typical field condition.

complex nature of the field condition. The significant differences in the lateral position offset and bearing angle deviation among the test runs indicated that the navigation precision needed to be further improved in rough field conditions.

3.4. Field validation test results

Fig. 14 shows a typical experiment path in the field validation test as compared to the reference path of the target navigation map. The experiment path had a similar overall trend to the reference path. The experiment path fluctuated slightly along the reference path, which was likely caused by the complicated environment of the paddy field. The larger error at the row ends was attributed to the fact that the turning radius of the robot was large than the row spacing.

The field validation test experienced an average lateral position offset of 5.93 cm (Fig. 15a), which was 133 % higher than that of the row tracking test. The errors that occurred in the map-based navigation process were mostly responsible for the increased lateral position offset, which demonstrated that the appropriateness and correctness of the projection and route planning techniques were important to the overall navigation accuracy. On the contrary, the bearing angle deviation remained relatively unchanged in the field tests (Fig. 15b). In the field

validation test, the lateral position offset experienced notable variation among the test runs. It was possibly attributed to the complex field conditions, for example the wet soil surface, which would cause the lateral position error by affecting the driving force of the robot. The maximum lateral position offset measured was 6.8 cm in the field validation test, which was within the lateral row clearance of the robot (9.2 cm), indicating a high unlikelyhood of rice seedling damage in operation. However, the resulting lateral position error of the field validation test was higher than that simulated through the deep learning-based model developed by Silva et al. (2023), where the average displacement error was recorded at 3.65 cm. Zhang et al. (2020) developed a 2D laser automatic navigation system for standardized orchards. In their field test, the average deviation was 4.8 cm, which is better than the results of the field validation test. Compared with the simulation experiment and standardized orchard test, the complex conditions of the paddy field contributed to the larger average lateral position error. On the other hand, the field validation test results were in agreement with that of a transplanter experiment by Nagasaka et al., (2004) in terms of navigation accuracy. It is therefore concluded that the robot was able to autonomously guide itself with a high level of accuracy and precision using the planting map of a transplanter.

4. Conclusions

As an integral part of an agricultural robot, the navigation system directly affects the performance of the robot. The navigation error of the robot was enlarged by the inclination of the robot, which was caused by the irregular field surface in paddy fields. In this study, a map-based navigation system was proposed, where the robot navigation map was derived from the planting map of a transplanter. The navigation system coupled the data from the GNSS and the IMU using the Kalman filter. The navigation error was corrected by the inclination information of the robot from the IMU. A series of simulations and tests were performed to evaluate the performance of the integrated navigation system, including the line following test, rough road simulation and test, row tracking test, and field validation test. The lateral position offset and bearing angle deviation were measured and analyzed to evaluate the performance. The robot had an average lateral position offset of 1.53 cm and an average bearing angle deviation of 4.36° in the line following test on a leveled concrete surface. The rough road test results showed that the integrated navigation system significantly reduced the lateral position offset and bearing angle variation by 38.3 % and 26.7 % respectively as compared to the control group without the Kalman filtering. In a typical paddy field condition, the average lateral position offset was 2.54 cm and the

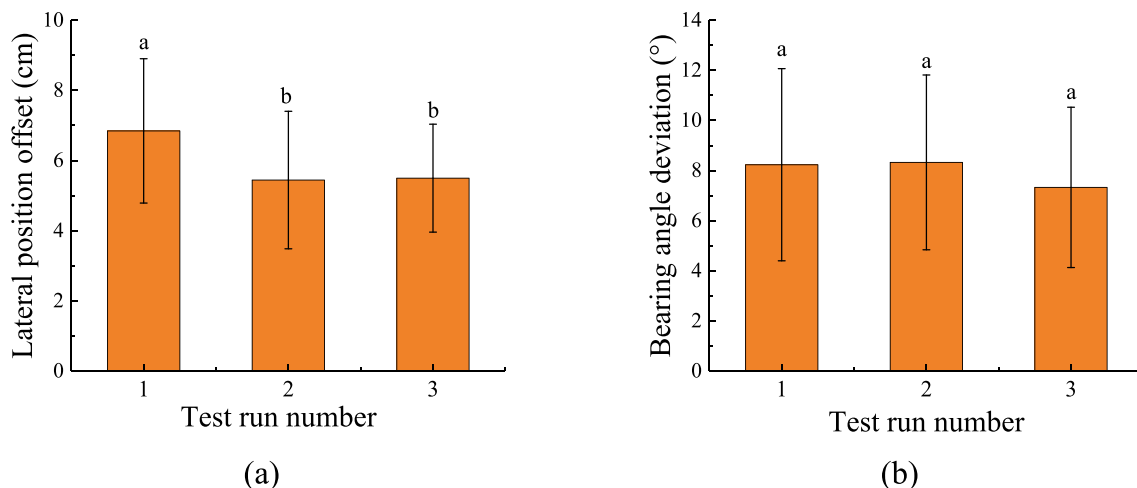


Fig. 15. Lateral position offset (a) and bearing angle deviation (b) of different test runs. Means followed by different letters are significantly different according to Tukey's multiple range test at the significance level of 0.05, and error bars are standard deviations.

average bearing angle deviation was 7.78°. Finally, in the field validation test, the average lateral position offset was 5.93 cm, which was deemed accurate enough for paddy field scouting operation. The integrated navigation system could be readily applied to the paddy field scouting robot to reduce the error of the navigation system.

There were several limitations to the proposed navigation system. Firstly, only two drive wheels were considered in developing the kinematic model of the robot and the supporting driven wheels were excluded for simplification purposes. Secondly, the frequency of the inclination was not accounted for in the system. A fixed collection frequency was applied in the integrated navigation system. However, in real-world field conditions, the inclination of the scouting robot is irregular, which could affect the precision of the system. Future studies could take the inclination frequency into account aiming to further decrease the navigation error of the system. Last but not least, the mechanism and affecting factors of the navigation system of the scouting robot were not fully examined, such as the slippage of the robot, due to the applied nature of this study. In a further step, the underlying mechanism of the robot movement should be thoroughly investigated to gain a better understanding of the principles of navigation accuracy improvement.

CRedit authorship contribution statement

Yuyuan Tian: Conceptualization, Methodology, Writing – original draft. **Zhenpeng Mai:** Methodology, Writing – review & editing. **Zhiwei Zeng:** Supervision, Project administration, Writing – review & editing. **Yinghu Cai:** Investigation, Visualization. **Jinpeng Yang:** Investigation, Visualization. **Bo Zhao:** Project administration, Visualization. **Xuhua Zhu:** Project administration, Visualization. **Long Qi:** Supervision, Funding acquisition, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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